Computer Organization & Architecture - Pipelining & Memory Hi

Course: Computer Architecture | Semester 4 | Contributor: Imlirenla Jamir

5-stage RISC pipeline, structural data and control hazards, forwarding, and cache mapping.

Pipeline Execution & Performance

* Standard 5 stages: Instruction Fetch IF, Decode ID, Execute EX, Memory MEM, Writeback WB.
* Speedup calculation: S = n * k / k + n - 1 approaching k as n becomes large.
* Structural hazards resource conflicts and solutions with separated I-cache and D-cache.

Data Hazards & Cache Organization

* RAW Read-After-Write hazards, pipeline stalls, and hardware forwarding bypassing paths.
* Direct-mapped, Fully Associative, and N-Way Set-Associative cache mapping formulas.
* Cache hit time, miss penalty, and write-through vs write-back policies.

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